



1 Methodology

Biofilm biomass was quantified by crystal violet staining (OD570) and normalized to bacterial growth (OD600). Specific Biofilm Formation (SBF) Index was calculated as: $SBF = (OD570 \text{ sample} - OD570 \text{ blank}) / (OD600 \text{ sample} - OD600 \text{ blank})$. Wells were grouped by strain/condition and media condition, with blank subtraction performed per matching media condition.

Classification thresholds applied — Weak biofilm: $SBF \geq 0.5$; Moderate biofilm: $SBF \geq 0.75$; Strong biofilm: $SBF \geq 1.1$; below the weak threshold, wells were classed Non-biofilm.

2 Instrument & acquisition details

OD600 file — Device: — Wavelength: nm Date: 08 May 2026 Time: 11:02:40 AM Temp: —

OD570 file — Device: — Wavelength: nm Date: 08 May 2026 Time: 12:15:51 PM Temp: —

3 Summary statistics

Strains/Conditions analysed: 23 Media conditions: 1 Blank wells: 4

Strong biofilm: 6 Moderate biofilm: 5 Weak biofilm: 1 Non-biofilm: 11

4 SBF Index by strain/condition — Figure 1



BiofilmQ-HT · SBF Biofilm Assay Report

Specific Biofilm Formation · crystal violet (OD570) normalized to growth (OD600)

Generated 08 Aug 2026

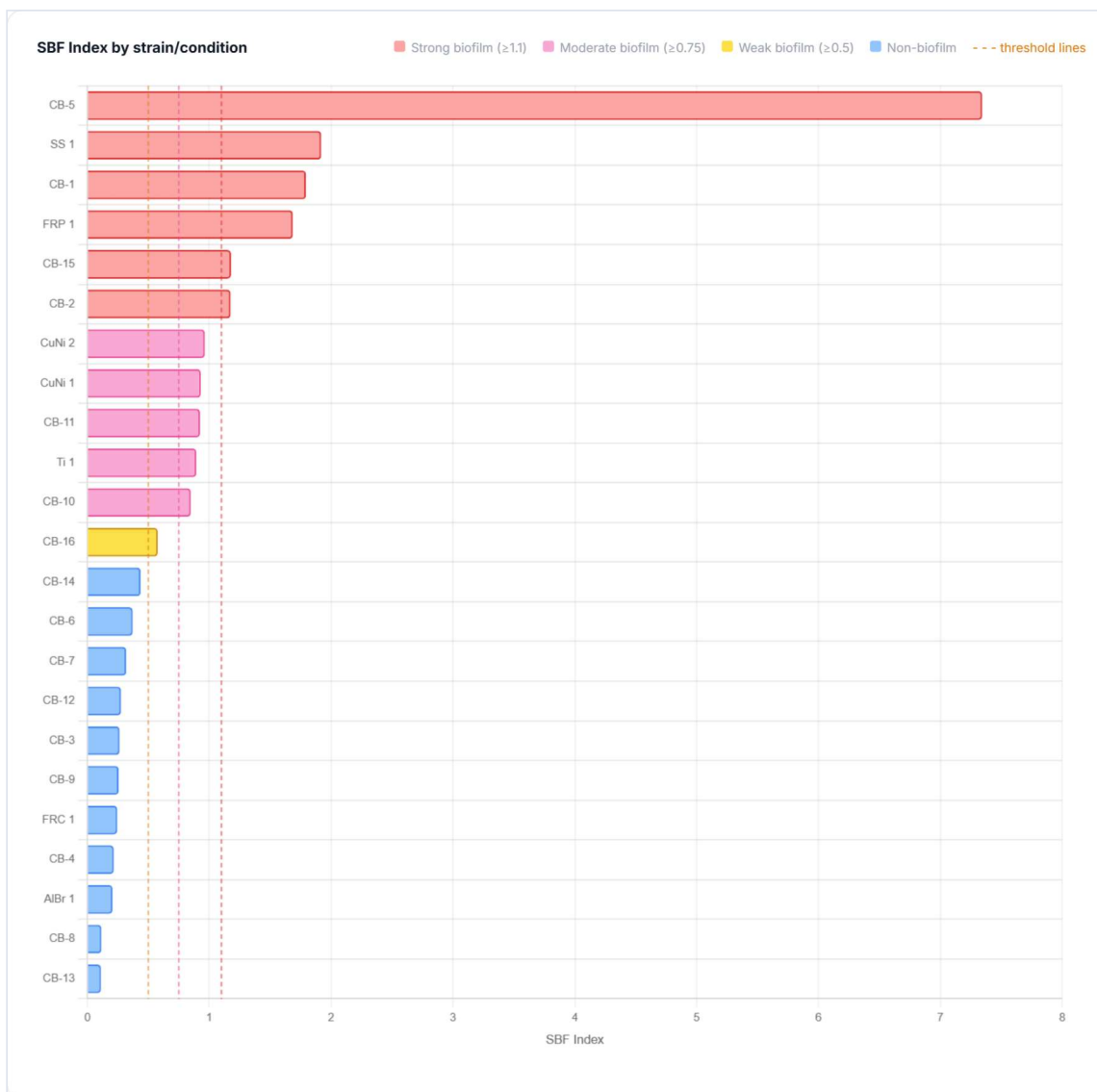




Figure 1. SBF Index per strain/condition, colour-coded by classification. Dashed lines mark the weak/moderate/strong thresholds.

5 Detailed results table — Table 1

Strain/Condition / ID	Type	Media	n	Mean OD600	Mean OD570	Blank OD600	Blank OD570	SBF Index	% Inhib.	Classification
CB-5	Strain/Condition	Default	4	0.321	2.148	0.045	0.119	7.343	—	Strong
SS 1	Strain/Condition	Default	4	1.349	2.622	0.045	0.119	1.919	—	Strong
CB-1	Strain/Condition	Default	4	0.528	0.986	0.045	0.119	1.794	—	Strong
FRP 1	Strain/Condition	Default	4	1.465	2.512	0.045	0.119	1.685	—	Strong
CB-15	Strain/Condition	Default	4	1.254	1.543	0.045	0.119	1.178	—	Strong
CB-2	Strain/Condition	Default	4	1.016	1.260	0.045	0.119	1.175	—	Strong
CuNi 2	Strain/Condition	Default	4	1.463	1.487	0.045	0.119	0.964	—	Moderate
CuNi 1	Strain/Condition	Default	4	0.430	0.478	0.045	0.119	0.930	—	Moderate
CB-11	Strain/Condition	Default	4	1.394	1.366	0.045	0.119	0.924	—	Moderate
Ti 1	Strain/Condition	Default	4	1.483	1.404	0.045	0.119	0.894	—	Moderate
CB-10	Strain/Condition	Default	4	1.386	1.256	0.045	0.119	0.848	—	Moderate
CB-16	Strain/Condition	Default	4	1.590	1.011	0.045	0.119	0.577	—	Weak
CB-14	Strain/Condition	Default	4	0.616	0.369	0.045	0.119	0.437	—	Non-biofilm
CB-6	Strain/Condition	Default	4	0.276	0.205	0.045	0.119	0.371	—	Non-biofilm
CB-7	Strain/Condition	Default	4	0.191	0.166	0.045	0.119	0.319	—	Non-biofilm
CB-12	Strain/Condition	Default	4	1.443	0.505	0.045	0.119	0.276	—	Non-biofilm
CB-3	Strain/Condition	Default	4	0.328	0.194	0.045	0.119	0.264	—	Non-biofilm
CB-9	Strain/Condition	Default	4	1.392	0.465	0.045	0.119	0.257	—	Non-biofilm
FRC 1	Strain/Condition	Default	4	1.471	0.469	0.045	0.119	0.245	—	Non-biofilm



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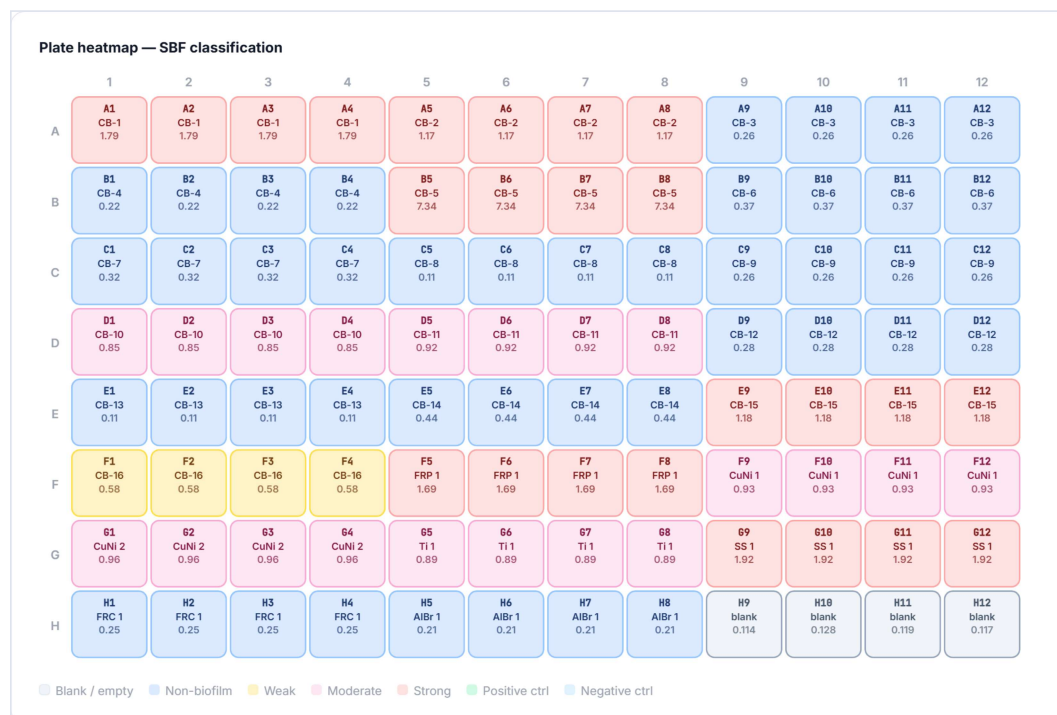
Specific Biofilm Formation · crystal violet (OD570) normalized to growth (OD600)

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Strain/Condition / ID	Type	Media	n	Mean OD600	Mean OD570	Blank OD600	Blank OD570	SBF Index	% Inhib.	Classification
CB-4	Strain/Condition	Default	4	0.121	0.136	0.045	0.119	0.216	—	Non-biofilm
AlBr 1	Strain/Condition	Default	4	0.694	0.253	0.045	0.119	0.206	—	Non-biofilm
CB-8	Strain/Condition	Default	4	0.289	0.147	0.045	0.119	0.115	—	Non-biofilm
CB-13	Strain/Condition	Default	4	1.451	0.275	0.045	0.119	0.111	—	Non-biofilm

Table 1. Per-strain/condition results, blank- and media-matched. n = number of replicate wells averaged.

6 Plate heatmap — Figure 2





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Figure 2. 96-well plate layout coloured by classification, with per-well SBF index shown.